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Thursday Commissioning Simulation — Instructor Pack

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Classification: Internal

Thursday Commissioning Simulation — Instructor Pack

Confidential — instructor and simulator operator only.

This pack covers the Thursday simulation day. Teams swap operator/ observer roles relative to Wednesday. Day target is **First Light** on the multispectral imager.

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1. Day at a glance

When	What	Mode
09:00 – 09:30	Standup, scenario brief to both teams	Instructor-led
09:30 – 12:00	**Pass-set 1** — Team A operates two compressed passes; Team B observes	Commissioning sim
12:00 – 13:00	Lunch + observer prep	—
13:00 – 13:15	Reset + brief for Pass-set 2	Instructor-led
13:15 – 15:45	**Pass-set 2** — Team B operates two compressed passes; Team A observes	Commissioning sim
15:45 – 17:00	Joint debrief, observer presentations, anomaly tickets	Plenary + individual

Each pass-set covers ~4 simulated ground passes in 2.5 h real time using a compressed-time multiplier (3×–5× between passes, 1× during passes). The simulator operator switches the speed by hand (§2).

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2. Scenario configuration

- **Breakpoint:** BP-3 — Commissioning Day Start (NEW; spec in `breakpoint_spec.md`).
- **Phase at start:** 4 (LEOP-complete).
- **Initial visibility:** Iqaluit, AOS at scenario T+0.
- **Planned passes during shift:** 4 (Iqaluit, Troll, Iqaluit, Troll — actual visibility depends on orbit).
- **Speed multiplier:** instructor-controlled; defaults to 3× between passes, 1× during passes.

The only start-up flag is `--speed` (default 1.0). During the shift, change the speed from the instructor console's speed buttons (0.5×, 1×, 2×, 5×, 10×) or over the instructor API, e.g. `{"type": "set_speed", "value": 3}` posted to `/api/command` for a 3× setting the buttons do not offer. `simulation.yaml` clamps any request to 1000×, and the simulator holds ≤ 10× while a burn is firing. The simulator does **not** change speed on its own around passes: the simulator operator does it by hand, using the contact schedule.

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3. Procedure expectations

The team is expected to drive commissioning in approximately this order across the four passes:

Pass 1. COM-001 EPS Checkout · COM-002 TCS Verification · COM-004 AOCS Actuator Checkout (wheels enabled, NOMINAL commanded) · COM-003 AOCS Sensor Calibration (gyro calibration in that NOMINAL, ruling Q16) · COM-005 AOCS Mode Transitions · COM-006 TTC Link Verification.

Pass 2. COM-007 OBDH Checkout · COM-008 FDIR Configuration (enable S12 monitoring, verify the S12/S19 rule sets).

Pass 3. COM-009 Payload Power On · COM-010 FPA Cooler Activation (this runs through the eclipse and into pass 4 — heat sink during eclipse, cool-down on the day side).

Pass 4. COM-011 Payload Calibration · **COM-012 First Light** — acquire one image of an ocean-current target listed in `planning/imaging_targets.yaml`.

End-of-shift target state: phase 5+ (NOMINAL ready), payload calibrated, one usable image acquired.

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4. Anomaly injection menu

Pick one primary and one optional secondary per team. Injection files live in `~/SpaceMissionSimulation/configs/eosat1/scenarios/`.

Severity	Scenario file	What students see	Recovery
Easy	`eps_overcurrent.yaml` (line=`payload`)	Overcurrent on payload power-on	CTG-012 Overcurrent Response
Easy	`gs_rf_degradation.yaml`	BER climbs mid-pass	CTG-014 BER Anomaly
Medium	`fpa_overtemp.yaml`	FPA cooler can't hold its −15 °C setpoint	CTG-004 Thermal Exceedance + payload safe
Medium	`payload_corrupt_image.yaml`	First image returns corrupt	CTG-015 Corrupted Image Recovery
Hard	`aocs_sensor_cascade.yaml`	Star tracker degrades; pointing budget broken	CTG-008 Star Tracker Failure → CTG-002 + defer First Light
Hard	`eps_progressive_load_shed.yaml`	Battery degradation forces progressive load-shed during cooler activation	CTG-001 Under-Voltage Load Shed
Very hard	`aocs_wheel_failure.yaml` followed by `transponder_failure.yaml`	Wheel locks late in pass 3, then transponder fails next pass — multi-anomaly response	CTG-007 → CTG-003

Cue cards for six of the seven are in the instructor book, chapter 18: `eps_overcurrent.yaml` §18.2 (inject on `payload`, `line_index=3`, during COM-009), `fpa_overtemp.yaml` §18.8, `payload_corrupt_image.yaml` §18.9, `aocs_sensor_cascade.yaml` §18.10, `eps_progressive_load_shed.yaml` §18.11 and the `aocs_wheel_failure.yaml` + `transponder_failure.yaml` pair §18.12. The seventh is below, in the same format.

`gs_rf_degradation.yaml` — Ground Station RF Chain Degradation (Easy)

- **Cue moment.** Start it after COM-006 has baselined the link margin, so the team has a reference. Its events fire at T+60 s (`high_ber` on the spacecraft side, magnitude 0.3, ramped over 120 s), T+180 s (ground LNA, +3 dB noise figure) and T+360 s (ground antenna mispoint, 3 dB). The file's briefing asks for the RF bridge in FRAME mode (`SMO_RF_MODE=FRAME ./start.sh`).
- **First indicator.** Eb/N0 (`ttc.eb_n0` , 0x0519) falls and BER (`ttc.ber` , 0x050C) rises on the TT&C display while the link margin (`ttc.link_margin` , 0x0503) erodes.
- **Recovery procedure.** CTG-014 BER Anomaly; CTG-003 if the link is lost.
- **Pass criterion.** The team compares RSSI (`ttc.rssi` , 0x0502) with Eb/N0 to separate the spacecraft-side degradation from the two ground-side ones before commanding anything on board, and files one ticket per cause.
- **Debrief Qs.** 1. Which reading told you part of the loss was on the ground? 2. Did anyone command the spacecraft to fix a ground fault? 3. When did TT&C first call the trend on the loop? 4. Was the HIGH-tier dump deferred or moved to another pass? 5. What would you check first next time?

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5. Pass operator-side criteria

Criterion	Pass-set 1 (Team A)	Pass-set 2 (Team B)
Phase advanced to COMMISSIONING (5)	✓	✓
EPS checkout complete with no faults	✓	✓
AOCS NOMINAL mode (4) achieved, pointing < 0.5°	✓	✓
FDIR rule sets enabled and verified (COM-008)	✓	✓
Payload powered ON	✓	✓
FPA cooler at its setpoint (−15 °C)	✓	✓
First Light image acquired (or formally deferred)	✓	✓
Anomaly tickets filed for all injections	✓	✓

The "formally deferred" option is genuine: if the team correctly diagnoses that pointing is insufficient and chooses to defer imaging, they pass the criterion. If they image anyway and produce garbage, they fail.

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6. Coaching during the shift

As the LEOP pack, §6: voice-loop discipline, procedure compliance, an FDIR-panel-level hint after five minutes stuck, and no pausing during a pass. Two commissioning additions:

- **Speed changes only between passes.** Compression is the simulator operator's call, never the team's, and the clock runs at 1× from AOS to LOS.
- **Handover quality between passes.** At each pass boundary the FD restates the open items, the procedure in progress and the next pass's goal; note anything dropped. Pass-to-pass continuity is harder than continuity within one pass, and First Light depends on it.

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7. Debrief

Same structure and timings as the LEOP pack, §7: hot-wash (15 min), anomaly walk-through (20 min), peer-review worksheet readout (20 min), lessons-learned synthesis (15 min) and the brief for Friday (5 min). The observer worksheets used today are *Team* and *Space usage*; Wednesday used *Communications* and *Technical*. The debrief prompts for each procedure are in textbook Part 2, §11.8.

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8. Safety, red lines, references

Safety and red lines are those of the LEOP pack, §8: no commanding without FD authorisation, no pausing the simulator to win an exercise (between-pass compression is set by the simulator operator, not the team), and no silent FD.

References are those of the LEOP pack, §9, plus:

- `configs/eosat1/scenarios/commissioning_day_start.yaml` (NEW, written by the instructor from the BP-3 spec; not in the shipped set)
- `configs/eosat1/procedures/commissioning/*.md`
- `configs/eosat1/mcs/role_analysis/payload_ops_role.md` (today is heavy on payload operations)
- Textbook Part 2, chapter 11 (commissioning operations), and the instructor book, chapter 18 (cue cards)